Systems Analysis and Design

Chapter 4: Designing architecture

Learning Objectives

4.1 Overview of architecture design

4.2 Types of IS architecture

4.3 Describe cloud computing and other current trends that help organizations address IS infrastructure-related challenges

4.4 Describe standards shaping the design of Internet based systems, options for ensuring Internet design consistency

Architectural Abstraction

- **Architecture in the small** is concerned with the architecture of individual programs.
  - At this level, we are concerned with the way that an individual program is decomposed into components.
- **Architecture in the large** is concerned with the architecture of complex enterprise systems that include other systems, programs, and program components.
  - These enterprise systems are distributed over different computers, which may be owned and managed by different companies

Why is architecture important?

- The architecture of a system is an abstract description of its structure and behavior
- The level of abstraction is chosen such that all critical design decisions are apparent and meaningful analysis is feasible

Architectural Representations

- Simple, informal block diagrams showing entities and relationships are the most frequently used method for documenting software architectures.
- But these have been criticized because they lack semantics, do not show the types of relationships between entities nor the visible properties of entities in the architecture.
- Depends on the use of architectural models.The requirements for model semantics depends on how the models are used.

Architectural Design Decisions

1. Generic application architecture that can be used?
2. How will the system be distributed?
3. What architectural styles are appropriate?
4. What approach will be used to structure the system?
5. How will the system be decomposed into modules?
6. What control strategy should be used?
7. How will the architectural design be evaluated?
8. How should the architecture be documented?

Architecture Reuse

- Systems in the same domain often have similar architectures that reflect domain concepts
- Application product lines are built around a core architecture with variants that satisfy particular customer requirements
- The architecture of a system may be designed around one of more architectural patterns or ‘styles’
- *A major goal of the software architecture discipline is to develop an environment that promotes the reuse of components across separate software projects (Ryan O'Farrell, J. A. Hamilton)*

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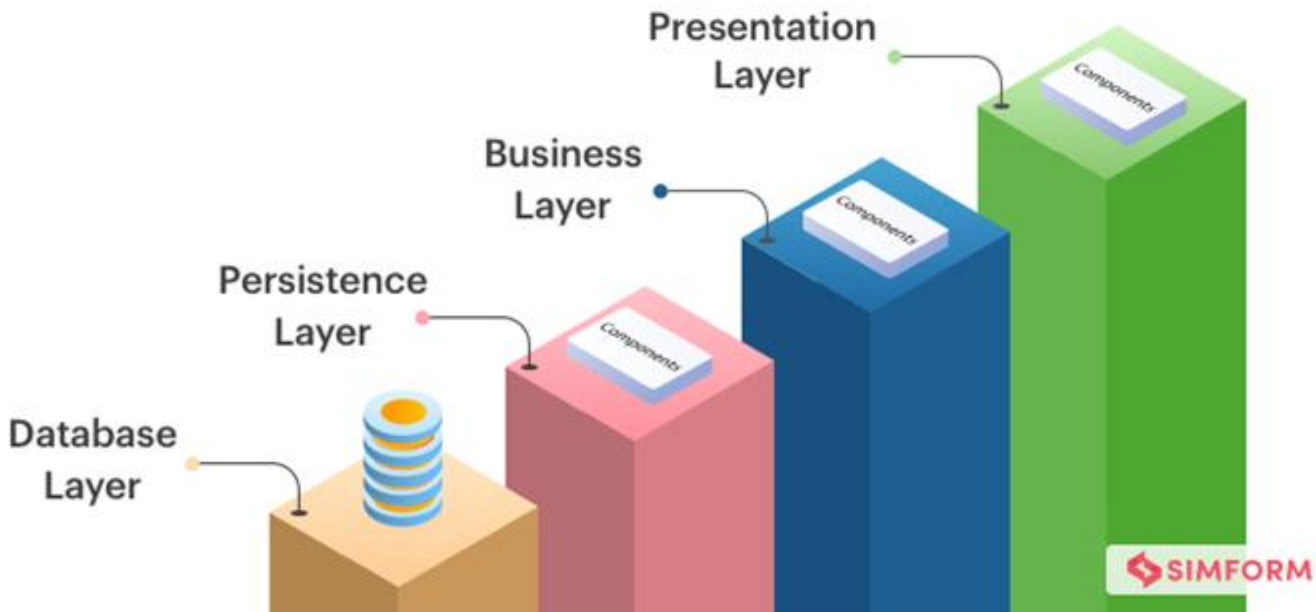
Many IS architectural styles

- Composition
  - static
    - layered
    - controller
    - client/server
    - pipes and filters
    - peer to peer
    - proxy / object-based
  - dynamic
    - object-oriented
    - software bus
    - microservices
- Communication
  - message passing
  - event driven
- Coordination
  - blackboard
  - shared tuple space
  - database/futures
- Emerging styles
  - mobile computing
  - sensor networks

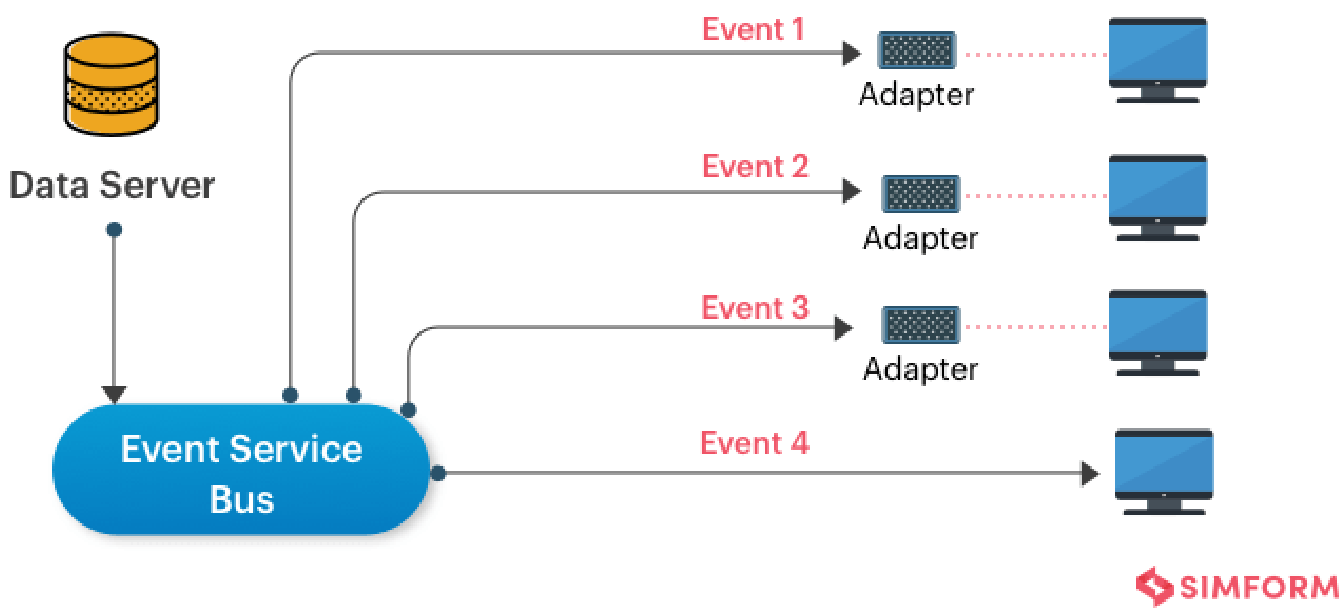
Layered Architecture Pattern

- Usage:
  - Applications that are needed to be built quickly.
  - Appropriate for teams with inexperienced developers and limited knowledge of architecture patterns.
  - Require strict standards of maintainability and testability



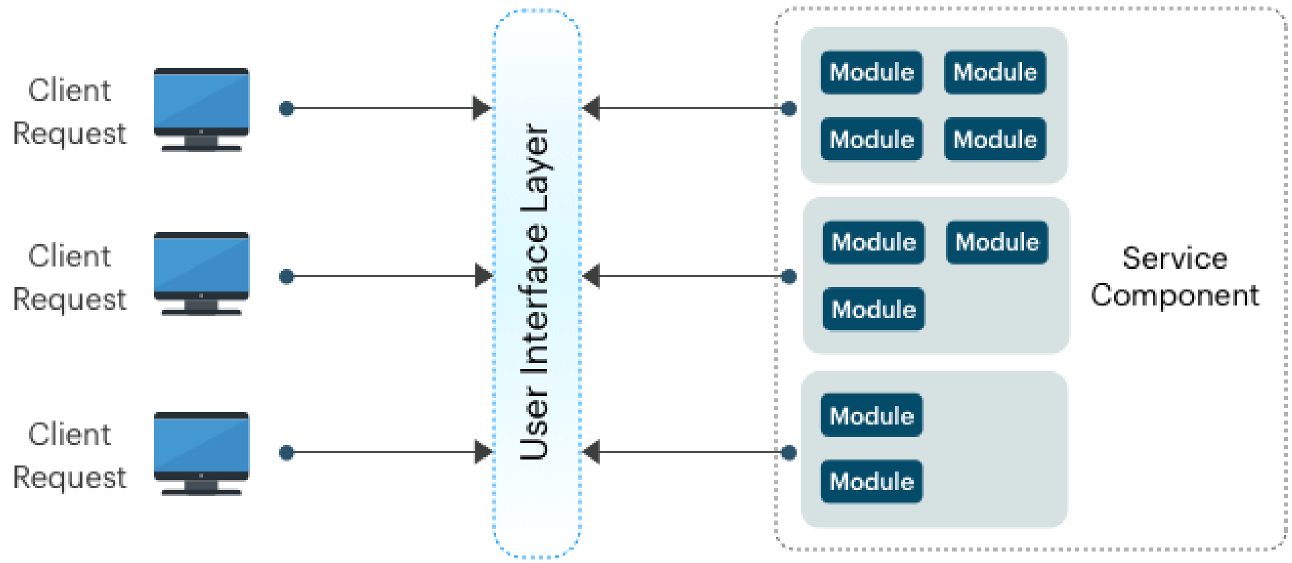
Event-driven Architecture Pattern

- Usage:
  - For applications where individual data blocks interact with only a few modules.
  - Helps with user interfaces



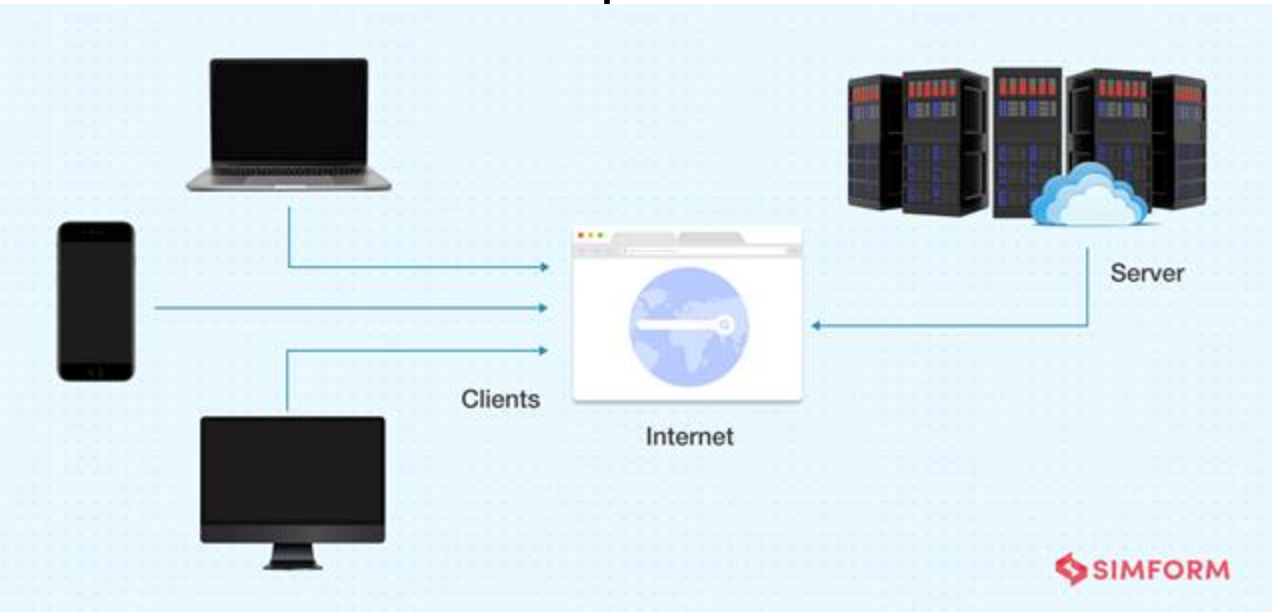
Microservices Architecture Pattern

- Usage:
  - Businesses and web applications that require rapid development.
  - Websites with small components, data centers with well-defined boundaries, and remote teams globally



Client-server Architecture Pattern

- Usage:
  - Applications that focus on real-time services
  - Applications that require controlled access and offer multiple services for a large number of distributed clients
  - An application with centralized resources and services that has to be distributed over multiple servers



Designing Systems for a Client/Server Architecture

- **Database engine** – (back-end) portion of the client/server database system running on the server that provides database processing and shared access functions
- **Client** – (front-end) portion of the client/server database system that provides the user interface and data manipulation functions
- **Application program interface (API)** – the software building blocks that are used to ensure that common system capabilities, such as user interfaces and printing, as well as modules are standardized to facilitate data exchange between clients and servers

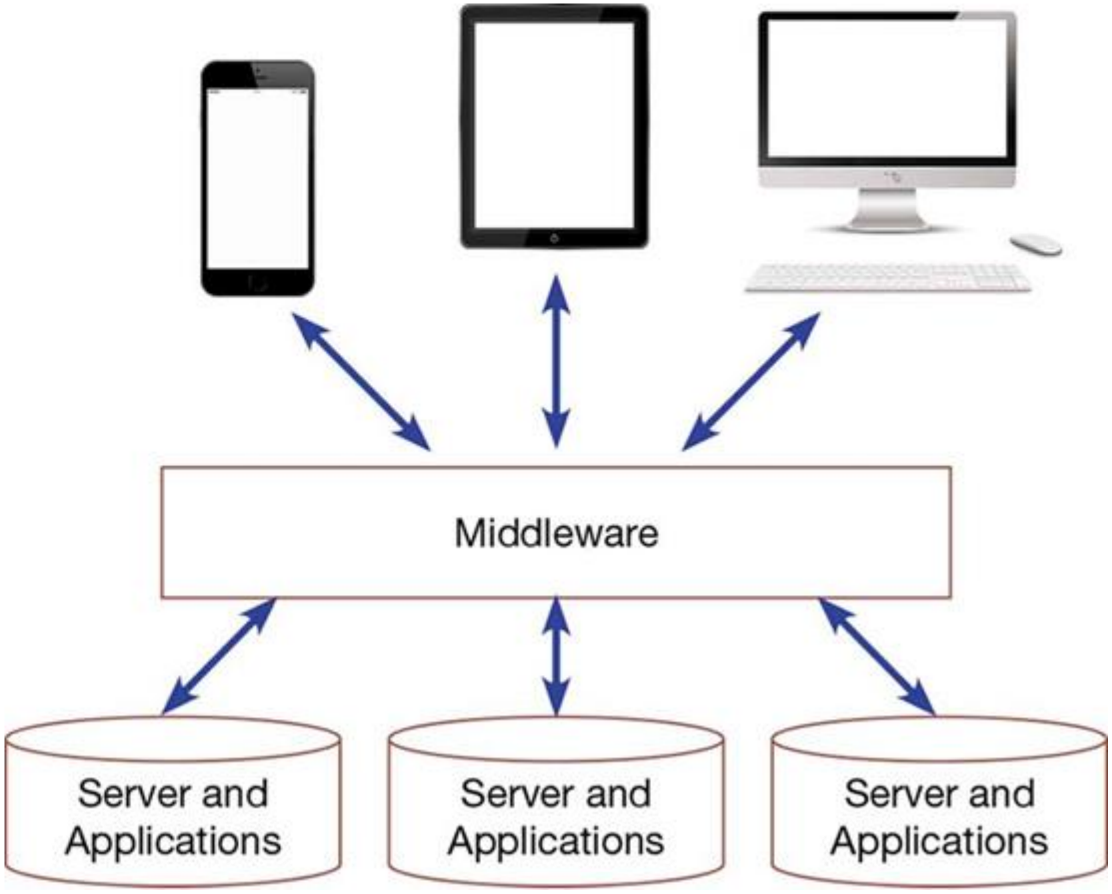
Advanced Forms of Client/Server Architectures (1 of 2)

- **Application server** – computing server where data analysis functions primarily reside
- **Virtual machine** – software emulation of a physical computer system, both hardware and operating system, that allows more efficient sharing of physical hardware resources
- **Virtualization** – act of creating virtual (rather than physical) versions of a variety of computing capabilities including hardware platforms, operating systems, storage devices, and networks

Advanced Forms of Client/Server Architectures (2 of 2)

- **Three-tiered client/server architecture** – advanced client/server architectures in which there are three logical and distinct applications—data management, presentation, and analysis—that are combined to create a single information system
- **Middleware** – combination of hardware, software, and communication technologies that brings data management, presentation, and analysis together into a three-tiered (or n-tiered) client/server environment

Middleware Ties Together Diverse Applications and Devices



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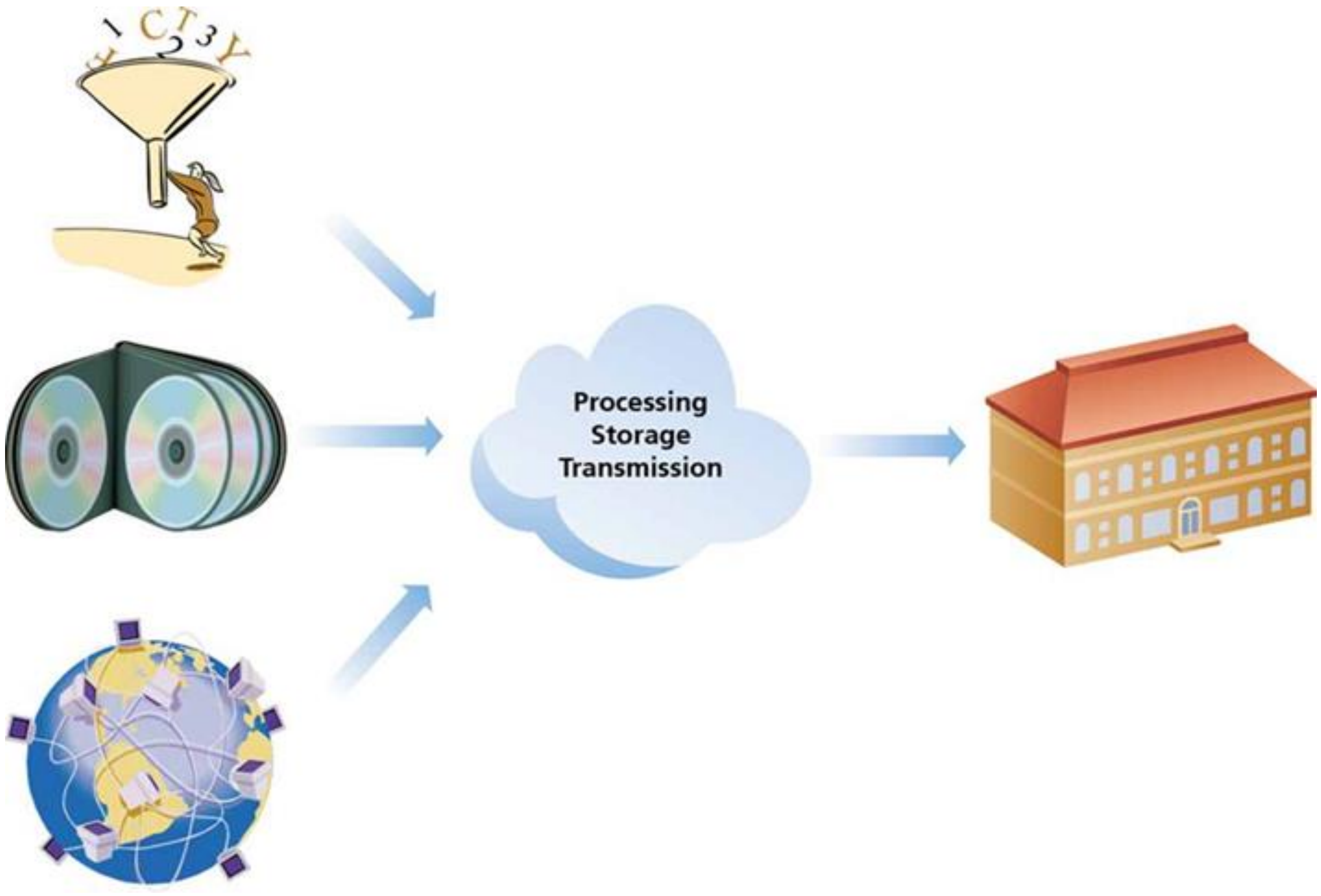
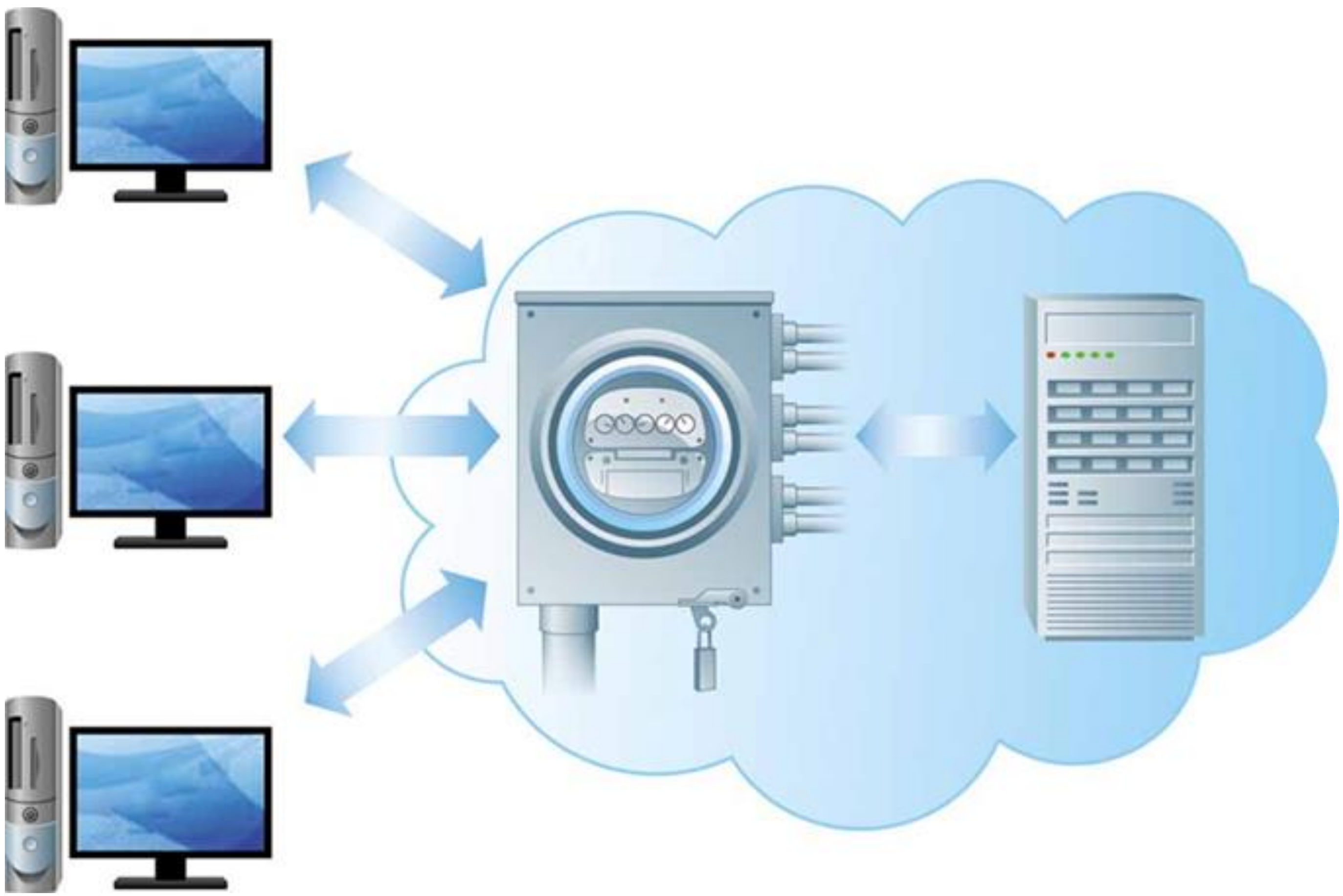
What is Cloud Computing?

- **Cloud computing** – the provision of applications over the Internet where customers do not have to invest in the hardware and software resources needed to run and maintain the applications, but are charged on a per-use basis
- **Utility computing** – form of on-demand computing where resources in terms of processing, data storage, or networking are rented on an as-needed basis. The organization only pays for the services used.

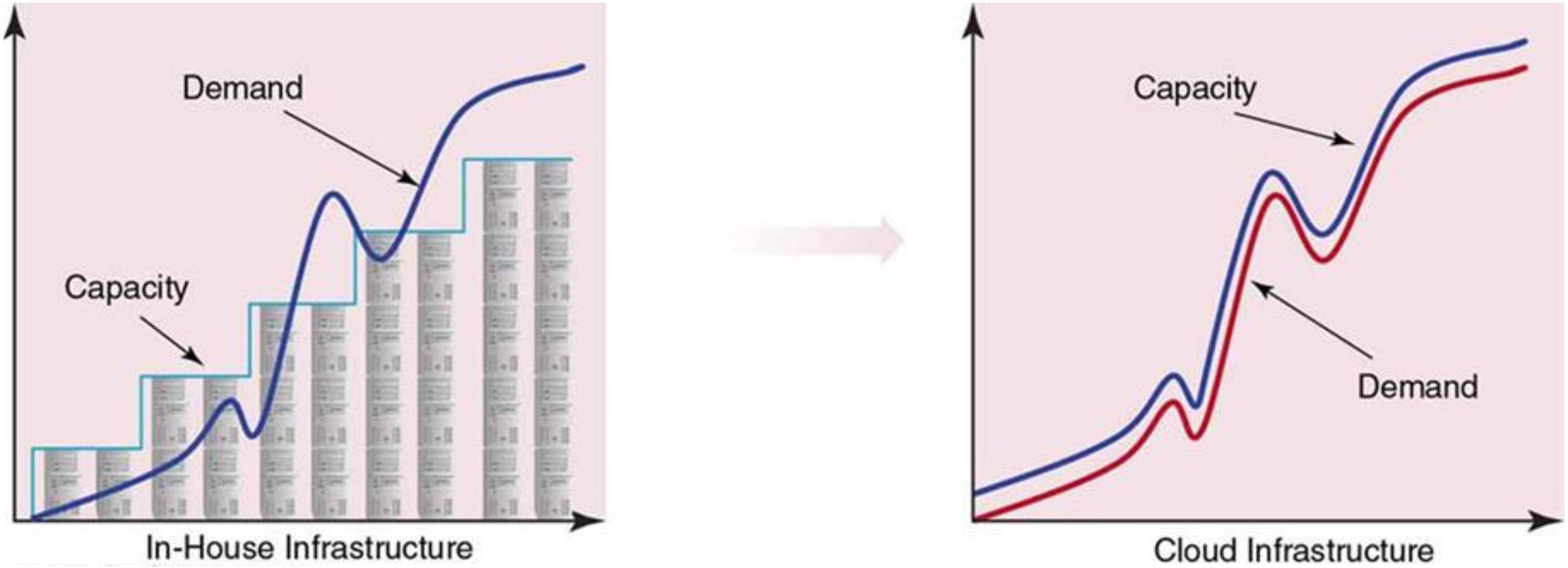
Processing, Storage, and Transmission of Data Taking Place in the Cloud



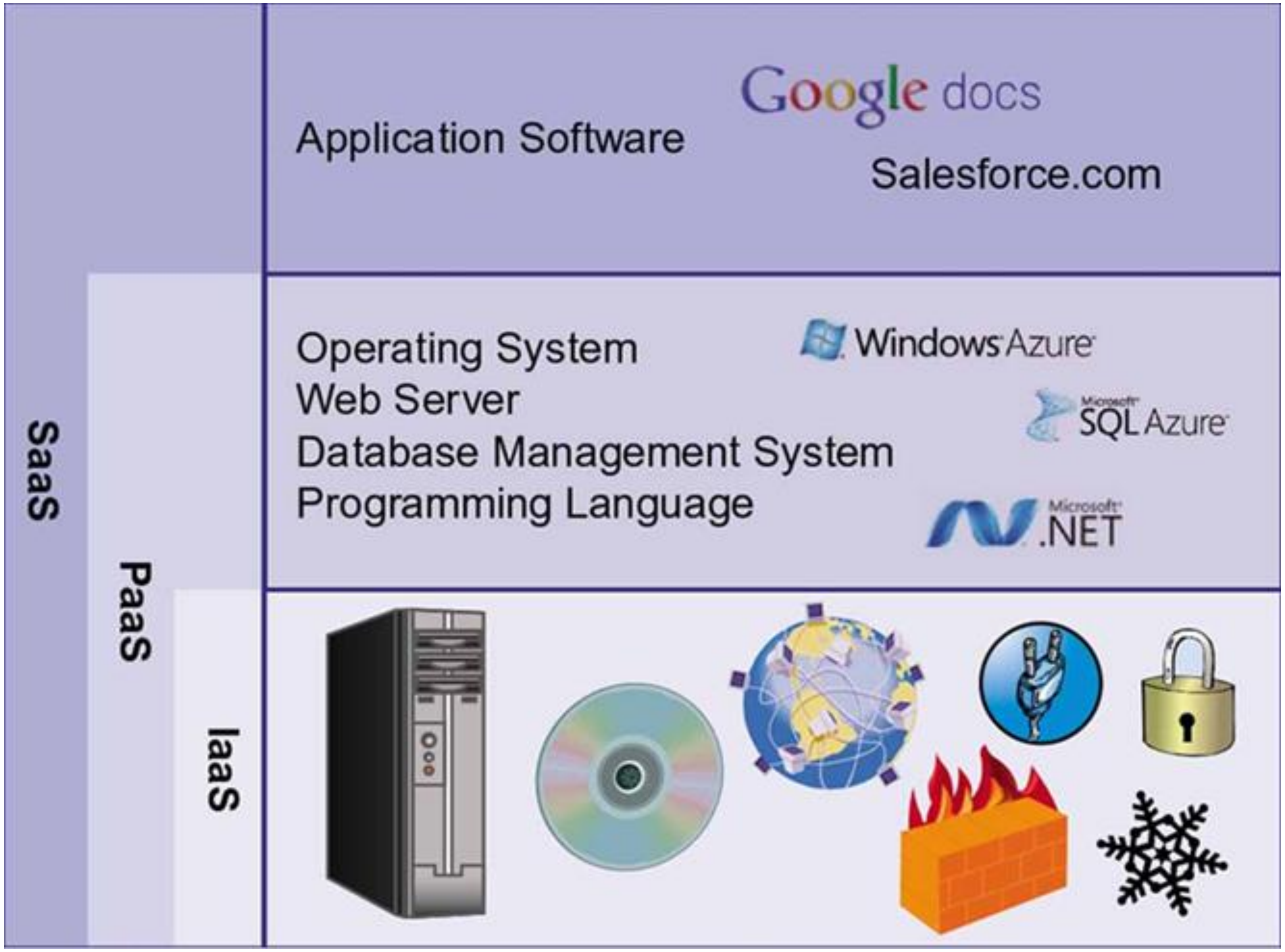
Cloud Computing Uses a Utility Computing Model, Allowing Companies to Pay for Computing Resources on an as Needed Basis



It Is Difficult to Match Demand Using an In-House Infrastructure; with a Cloud Infrastructure, Resources Can Be Added Incrementally, on an As-Needed Basis



Services by SaaS, PaaS, and IaaS Providers



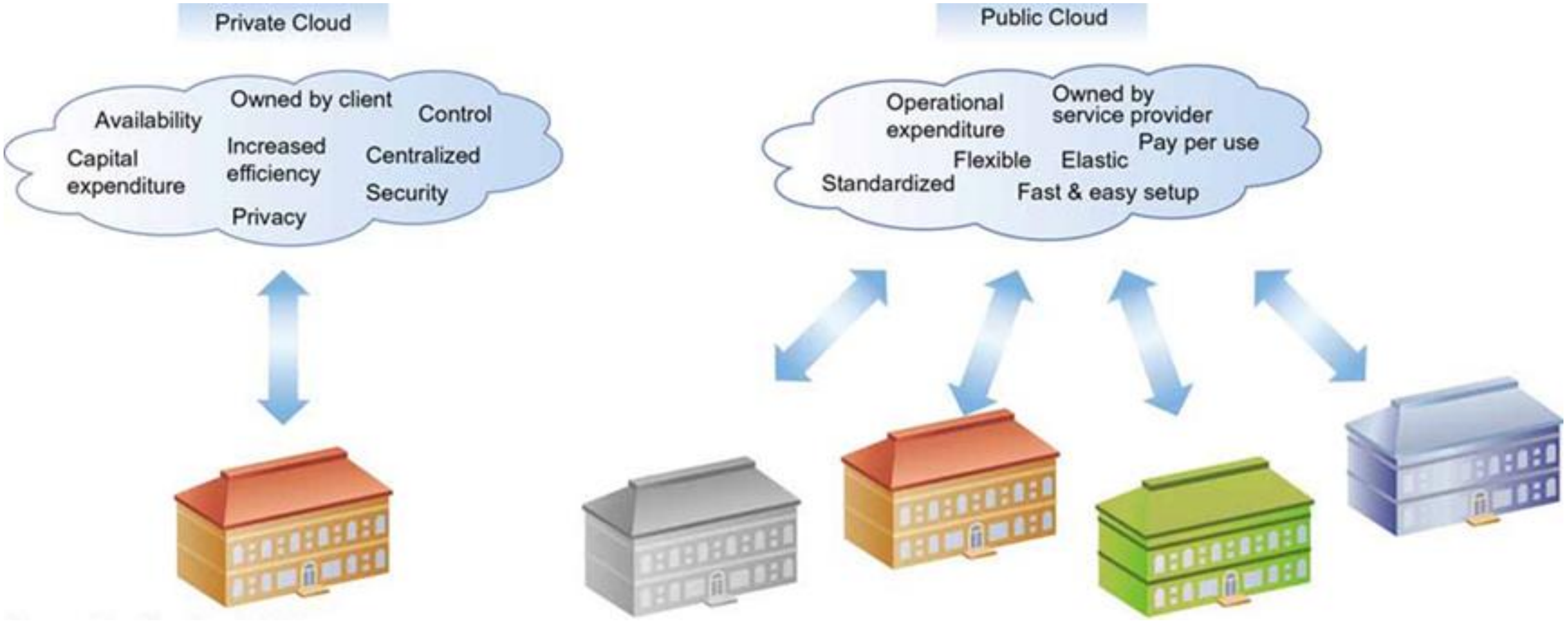
Service Models

- **Infrastructure as a service (IaaS)** – cloud computing model in which only the basic capabilities of processing, storage, and networking are provided
- **Platform as a service (PaaS)** – cloud computing model in which the customer can run his or her own applications that are typically designed using tools provided by the service provider; the customer has limited or no control over the underlying infrastructure
- **Software as a service (SaaS)** – cloud computing model in which a service provider offers applications via a cloud infrastructure

Cloud Characteristics

- Cloud computing has the following characteristics that distinguish it form in-house infrastructure:
  - On-demand self-service
  - Rapid elasticity
  - Broad network access
  - Resource pooling
  - Measured service

Public Clouds Versus Private Clouds

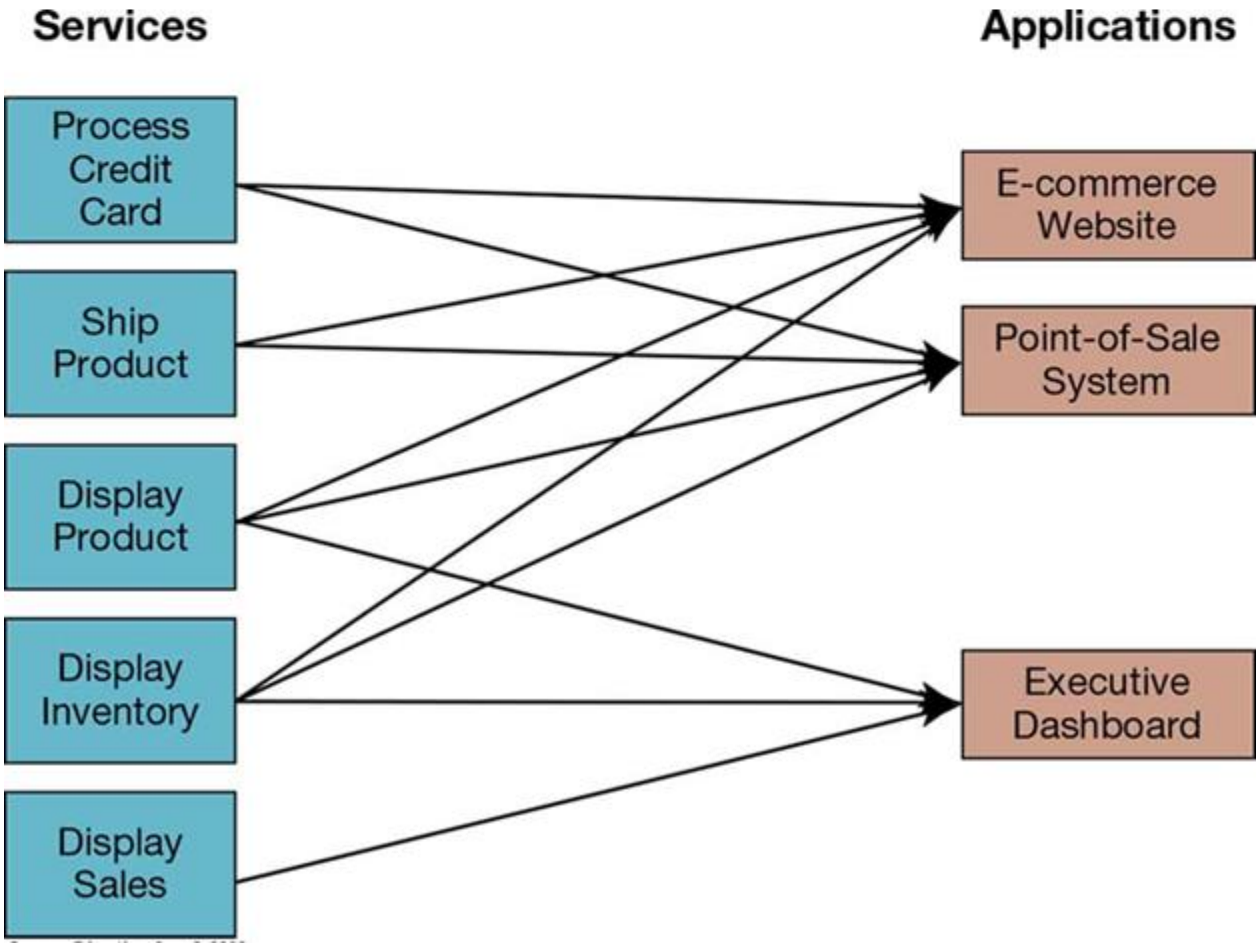


Organizations Have to Consider Various Issues When Managing Their Cloud infrastructure

Service-Oriented Architecture

- **Service-oriented architecture (SOA)** – software architecture in which business processes are broken down into individual components (or services) that are designed to achieve the desired results for the service consumer (which can be either an application, another service, or a person)
- Three main principles of SOA include:
  - **Reusability** – service should be usable in different applications
  - **Interoperability** – service should work with other service
  - **Componentization** – service should be simple and mobile

Using SOA, Multiple Applications Can Invoke Multiple Services



Web Services (1 of 2)

- **Web service** – method of communication between two electronic devices over a network
- **eXtensible Markup Language (XML)** – Internet authoring language that allows designers to create customized tags, enabling the definition, transmission, validation, and interpretation of data between applications
- **JavaScript Object Notation (JSON)** – lightweight data interchange approach that is relatively easy for humans to understand and for computers to generate or interpret

Web Services (2 of 2)

- **Simple Object Access Protocol (SOAP)** – protocol for communicating XML data between Web service applications and the operating system
- **Representational State Transfer (REST)** – relatively simple and fast protocol for communicating JSON data between Web service applications and the operating system

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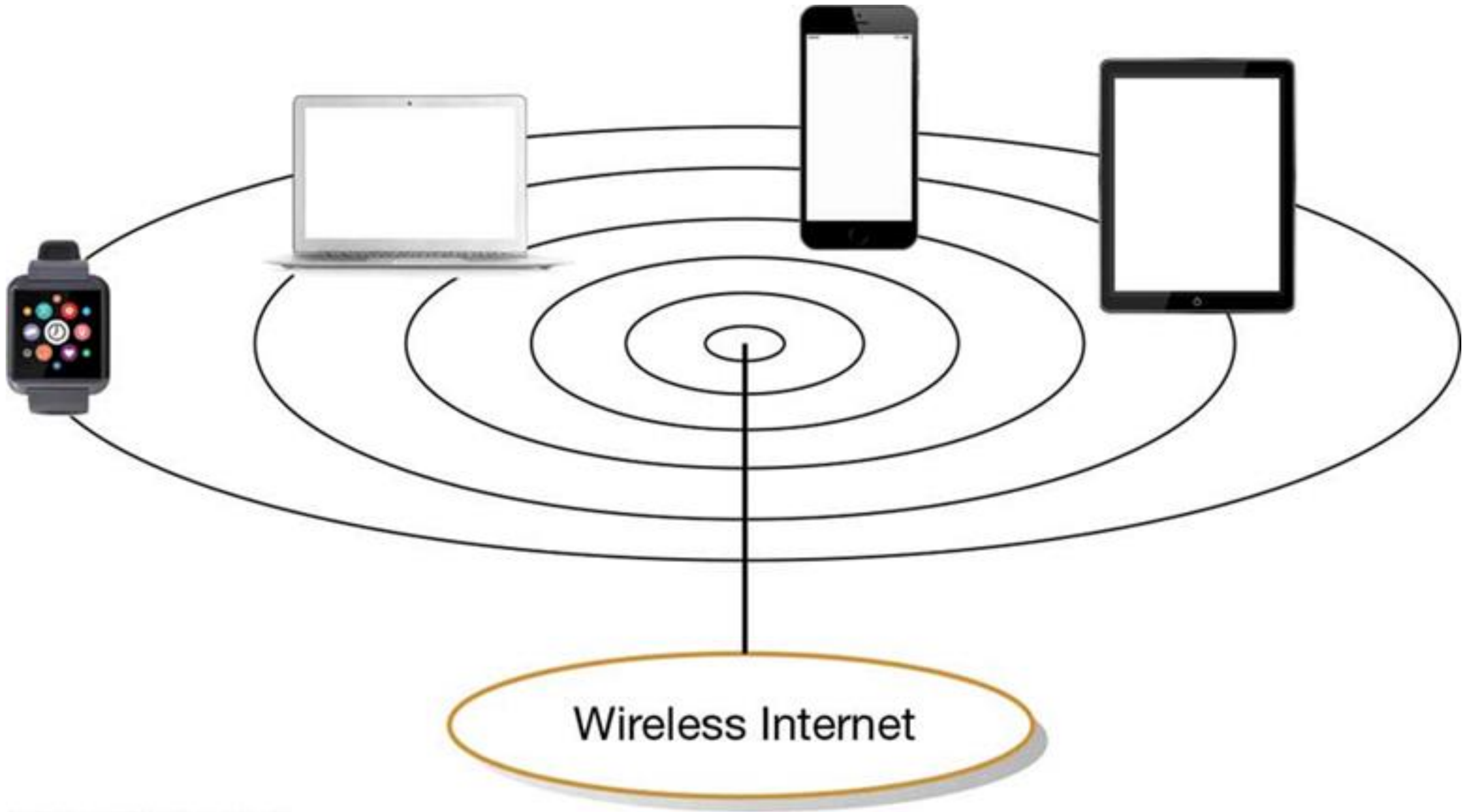
Designing Internet Systems

- Most new system development focuses on Internet-based applications (for internal processing, business-to-business, and business-to-consumer)
- Internet design is simpler than client/server due to proliferation of standards

Standards Drive the Internet

- Types of standards include:
  - **Domain naming system (BIND)** – method fortranslating Internet domain names into Internet Protocol (IP) addresses. BIND stands for Berkeley Internet Name Domain.
  - **Hypertext Transfer Protocol (HTTP)** – communication protocol or exchanging information on the Internet
  - **Hypertext Markup Language (HTML)** –standard language for representing content on the Web through the use of hundreds of command tags

Thin Clients Used to Access the Internet



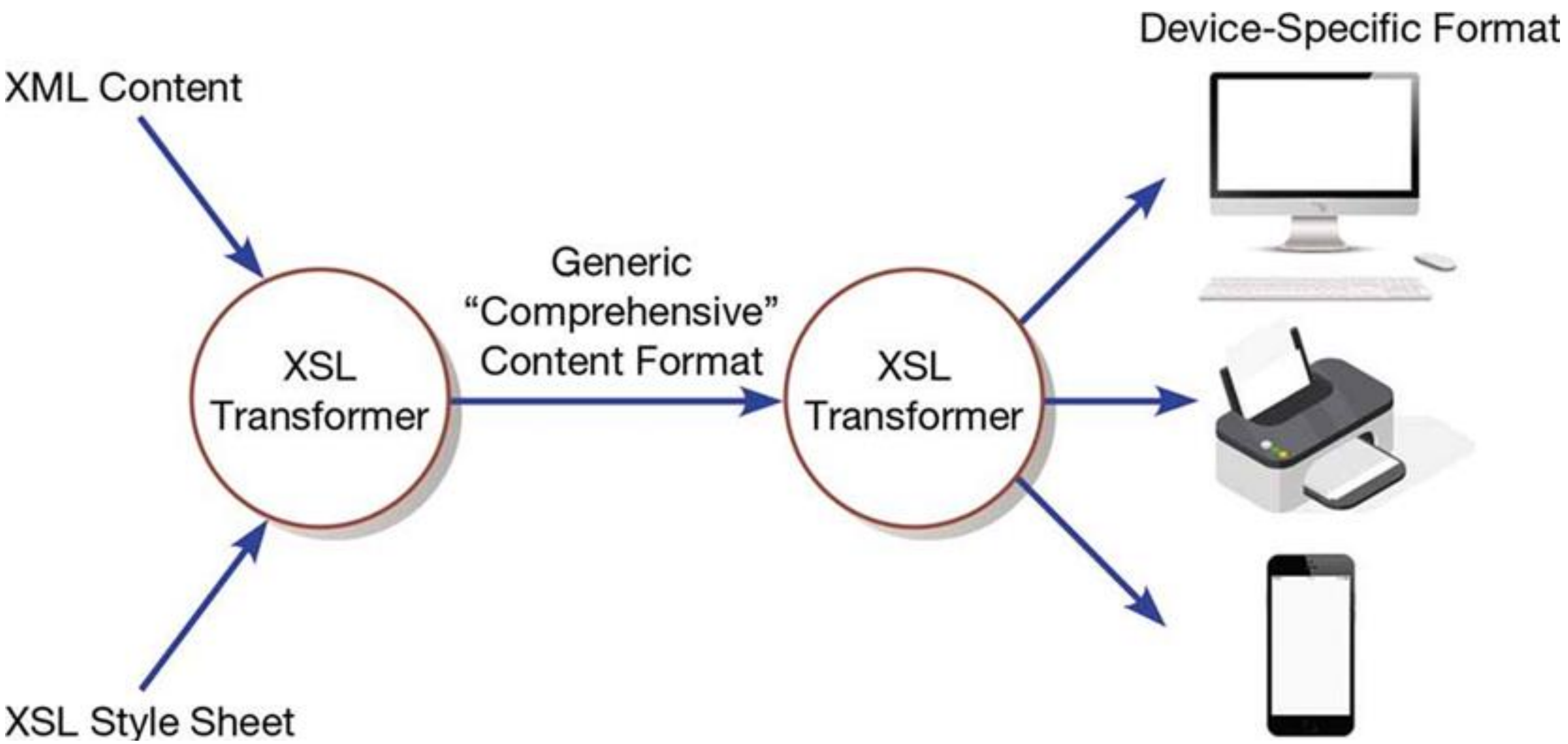
Site Consistency

- Professionalism requires a consistent look-and-feel across all pages of a website
  - Makes it easier for users to navigate
- **Cascading Style Sheets (CSSs)** –set of style rules that tells a Web browser how to present a document
- **eXtensible Stylesheet Language (XSL)** –specification for separating style from content when generating XML pages

Using HTML's Link Command for Cascading Style Sheets

Sample Command :	
LINK HREF="style5.css" REL=StyleSheet TYPE="text/css" TITLE="Common Back-ground Style" MEDIA="screen, print">	
Command Parameters:	
HREF="filename or URL"	Indicate the location of the linked object or document.
REL="relationship"	Specify the type of relationship between the document and linked object or document.
TITLE="object or document title"	Declare the title of the linked object or document.
TYPE="object to document type"	Declare the type of linked object or document.
MEDIA="type of media"	Declare the type of medium or media to which the style sheet will be applied (e.g., screen, print, projection, aural, braille, tty, tv, all).

Combining XML Data with XSL Style Sheet to Format Content



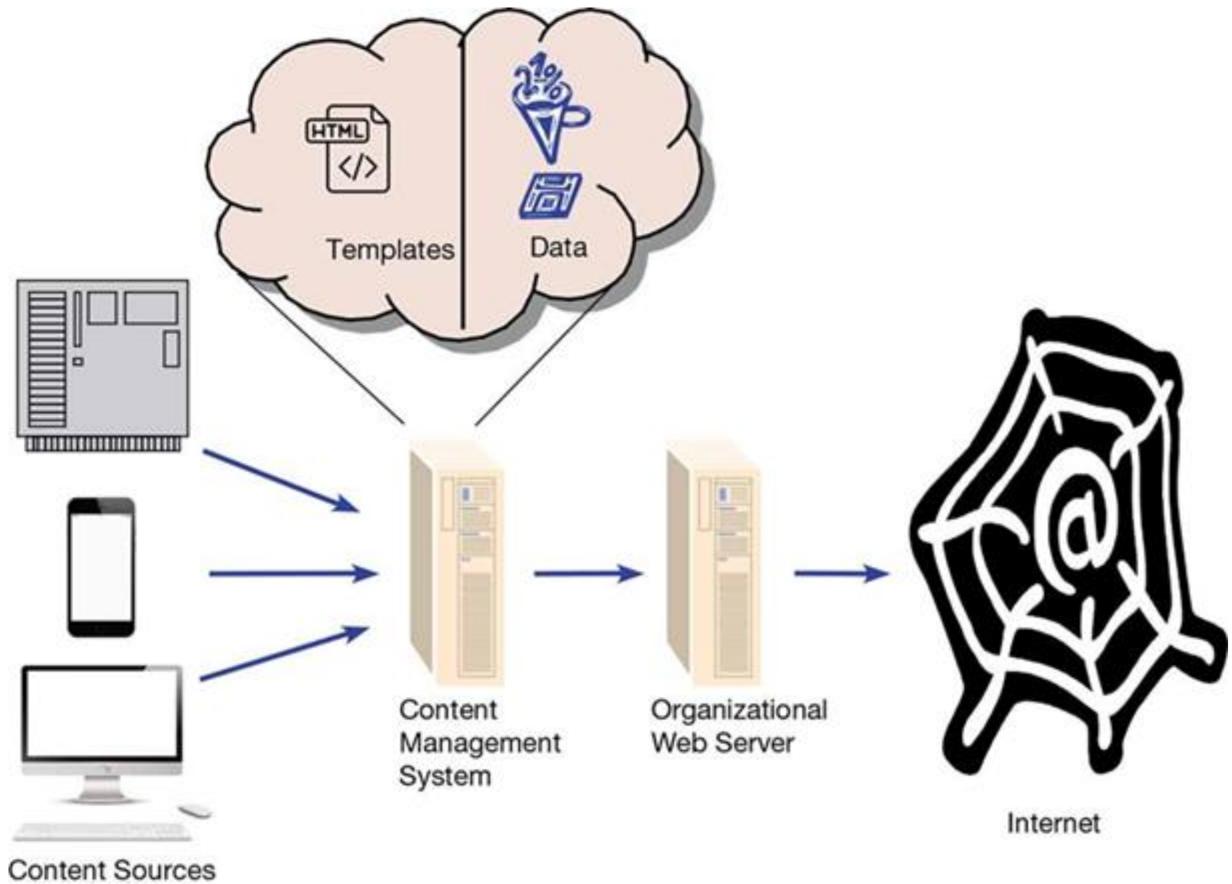
Design Issues Related to Site Management (1 of 2)

- Customer loyalty and trustworthiness results in customers feeling like the site, and their data, are secure. Ways to convey this:
  - **Design quality** (professional appearance)
  - **Up-front disclosure** (open and honest relationship)
  - **Comprehensive, correct, and current content** (provides users with current information)
  - **Connected to the rest of the Web** (credibility)

Design Issues Related to Site Management (2 of 2)

- **Personalization** – providing Internet content to a user based upon knowledge of that customer
- **Customization** – results in Internet sites that allow users to customize the content and look of the site based on their personal preferences

A Content Management System Allows Content from Multiple Sources to be Stored Separately From its Formatting to Ease Website Management



Summary

- In this chapter you learned to:
- Distinguish between file server and client/server environments, contrasting how each is used in a LAN
- Describe cloud computing and other current trends that help organizations address IS infrastructure-related challenges
- Describe standards shaping the design of Internet based systems, options for ensuring Internet design consistency